

A Catalogue of Topological Moiré Bands in Twisted Semiconductors

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Twisted two-dimensional semiconductors provide a route to flat and topological moiré minibands, but systematic principles for organizing their material dependence have remained unclear. Here, we establish a high-throughput framework that integrates structural relaxation, first-principles electronic structure calculations, and moiré band topology. We apply this framework to 43 experimentally realized monolayers and 91 symmetry-inequivalent bilayer prototypes, yielding over 1,000 angle-resolved moiré electronic band structures. This database reveals that the low-energy moiré electronic structure is organized primarily by the valley character of the parent band edge together with stacking symmetry. In Γ -valley systems, the miniband width usually follows a nearly quadratic twist-angle scaling, consistent with a folding-dominated kinetic-energy scale. In K -valley systems, stacking-controlled interlayer hybridization governs whether parent Berry curvature is redistributed into isolated valley Chern minibands. By contrast, M -valley systems form a more material-specific class associated with anisotropic and symmetry-constrained band folding. The same valley-and-stacking hierarchy rationalizes the emergence or suppression of $\mathbb{Z}_2$ minibands, and surface termination in Janus bilayers provides a microscopic knob for changing the relevant valley character. These results establish a materials-level organizing principle for designing flat and topological moiré bands in twisted semiconductors.

Twisted layered materials have emerged as a premier platform for engineering quantum matter through rotational misalignment, forming the foundation of twistronics [1–4]. In these systems, moiré potentials profoundly renormalize electronic states, enabling a rich landscape of emergent phenomena ranging from correlated insulators and unconventional superconductivity to topological quantum Hall effects [5–24]. This paradigm, initially centered on graphene [5, 10, 11], has rapidly expanded to transition metal dichalcogenides, hexagonal boron nitride, and low-dimensional magnets [7, 8, 12–31]. Yet, despite this remarkable versatility, it remains fundamentally unresolved whether universal organizing principles govern moiré electronic structures, or whether their behavior is intrinsically material-specific.

The lack of a predictive framework to discern universal scaling from material-dependent evolution stems from the extreme sensitivity of moiré bands to parent electronic structures, stacking symmetries, and twist angles [1, 3, 32, 33]. This impasse is largely a consequence of the prohibitive computational cost required to explore the expansive moiré parameter space with sufficient fidelity. At small twist angles, supercells containing thousands of atoms render standard first-principles calculations impractical [34]. Furthermore, the critical role of structural relaxation introduces atomic-scale reconstructions that profoundly reshape band topology, necessitating a

level of first-principles precision that is rarely scalable [19, 22, 24]. Consequently, systematic investigations remain restricted to a few prototypical cases, preventing a global understanding of the interplay between atomic structure and emergent physics.

In parallel, large-scale computational databases have transformed materials discovery by enabling the systematic identification of hidden regularities across high-dimensional spaces [35–43]. However, while recent efforts have initiated the exploration of twistable systems [19, 22, 44–46], these studies typically either survey a broad range of materials at a limited number of twist angles, or focus on specific material families at particular twist configurations, and therefore do not yet offer a systematically comparable dataset across a wide range of twist angles. The absence of such a foundation has hindered the quantitative identification of the organizing principles governing moiré electronic structure, particularly the complex interplay between valley character, twist angle, and topology. Addressing this gap requires a comprehensive, high-throughput approach to establish the fundamental rules that dictate the topological and electronic landscape across diverse moiré material platforms.

In this work, we bridge this gap by establishing a high-throughput computational framework and constructing a comprehensive database of twisted van der Waals bilayers, which integrates structural relaxation with high-fidelity electronic and topological characterization across an unprecedented range of materials, stacking configurations, and experimentally accessible twist angles. Our

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systematic analysis uncovers universal organizing principles, most notably a valley-controlled scaling of miniband bandwidths that quantifies the suppression of kinetic energy in moiré superlattices. Furthermore, we identify a valley-selective mechanism through which interlayer twisting induces a diverse array of topological minibands from trivial parent systems, stabilizing Chern and $\mathbb{Z}_2$ phases across broad twist-angle windows. These results establish a predictive, quantitative map connecting parent electronic structures to moiré-induced topology, providing a rigorous foundation for the rational design and analysis of twisted quantum matter.

COMPUTATIONAL WORKFLOW

To achieve this systematic exploration, we developed an integrated computational workflow organized into several distinct stages, each designed to ensure physical fidelity while navigating the expansive chemical space of twisted 2D materials (Fig. 1a).

Monolayer selection and screening. The process begins with a curated library of experimentally realized two-dimensional materials [41]. To focus on insulating and nonmagnetic systems, we apply a sequence of screening criteria that exclude metallic and magnetic monolayers. We further restrict the scope to materials with square or hexagonal symmetry, which facilitates the exhaustive enumeration of commensurate moiré supercells via twist indices [2, 47]. To maintain computational tractability at small twist angles, we constrain the primitive unit cell size, thereby defining a finalized candidate pool for subsequent bilayer assembly.

Bilayer construction and stacking enumeration. The selected monolayers are assembled into bilayer prototypes based on their intrinsic crystal symmetry. Using a group-theoretical approach, we identify all symmetry-inequivalent stacking configurations that arise from relative rotations and surface terminations. For centrosymmetric materials, this analysis identifies parallel (AA) and antiparallel (AB) stackings as the primary prototypes [22, 44]. In systems lacking mirror-related surfaces, additional configurations are considered to account for distinct termination combinations. For each prototype, we compute the generalized stacking-fault energy (GSFE) surface [48] by sampling interlayer translations, providing the requisite interlayer potential for the subsequent structural relaxation of twisted moiré superlattices.

High-accuracy structural relaxation and electronic characterization. Twisted structures are relaxed using a combination of GSFE-based continuum relaxation [48] and E(3)-equivariant machine-learning force fields (MLFFs) [51], enabling efficient optimization of large supercells with near first-principles accuracy [52]. Based on the relaxed geometries, electronic properties are evaluated using density functional theory to construct rep-

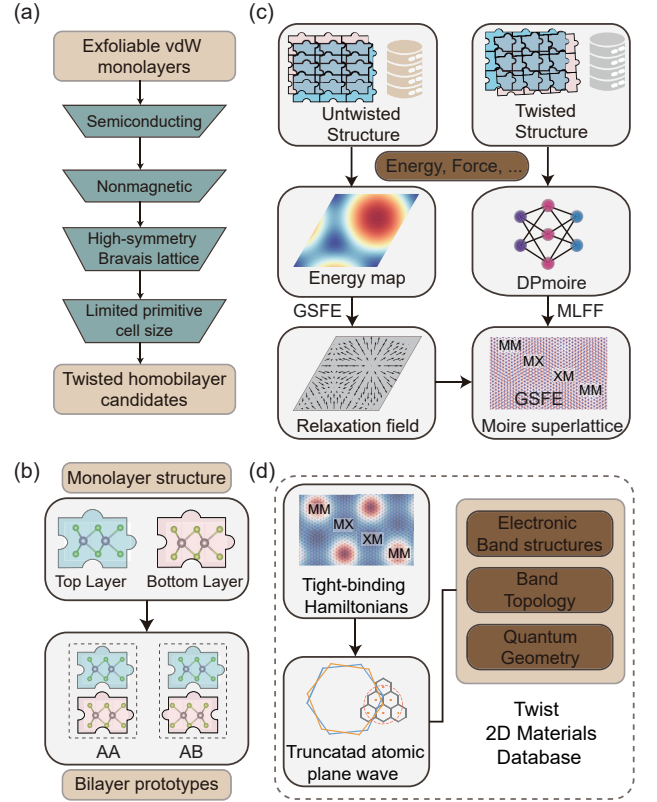


FIG. 1. Hierarchical high-throughput workflow for twisted two-dimensional materials. (a) An experimental 2D materials database serves as the input, from which candidate monolayers are selected using predefined screening criteria. (b) The screened monolayers are assembled into symmetry-inequivalent bilayer prototypes by enumerating distinct relative rotations and termination-dependent stacking configurations. (c) Interlayer sliding is sampled to construct the generalized stacking-fault energy (GSFE) surface, which provides initial stacking configurations for twisted structures. These are further relaxed using machine-learning force fields (MLFFs). (d) Starting from the relaxed structures, corresponding tight-binding Hamiltonians derived from first-principles calculations are constructed. The truncated atomic plane-wave (TAPW) method is then used to compute moiré electronic band structures and their topological properties, including band topology and quantum geometry. These results together constitute the twisted two-dimensional materials database.

resentative tight-binding Hamiltonians. The low-energy moiré minibands and their associated topological invariants, including Chern numbers and $\mathbb{Z}_2$ indices, are subsequently computed using the truncated atomic plane-wave (TAPW) method [44, 53, 54]. This computational pipeline allows for the high-throughput evaluation of electronic spectra across a wide range of twist angles, culminating in a comprehensive, angle-resolved database of moiré band structures, band topology and quantum geometry (Fig. 1d). This systematic foundation ultimately enables the identification of the universal organizing prin-

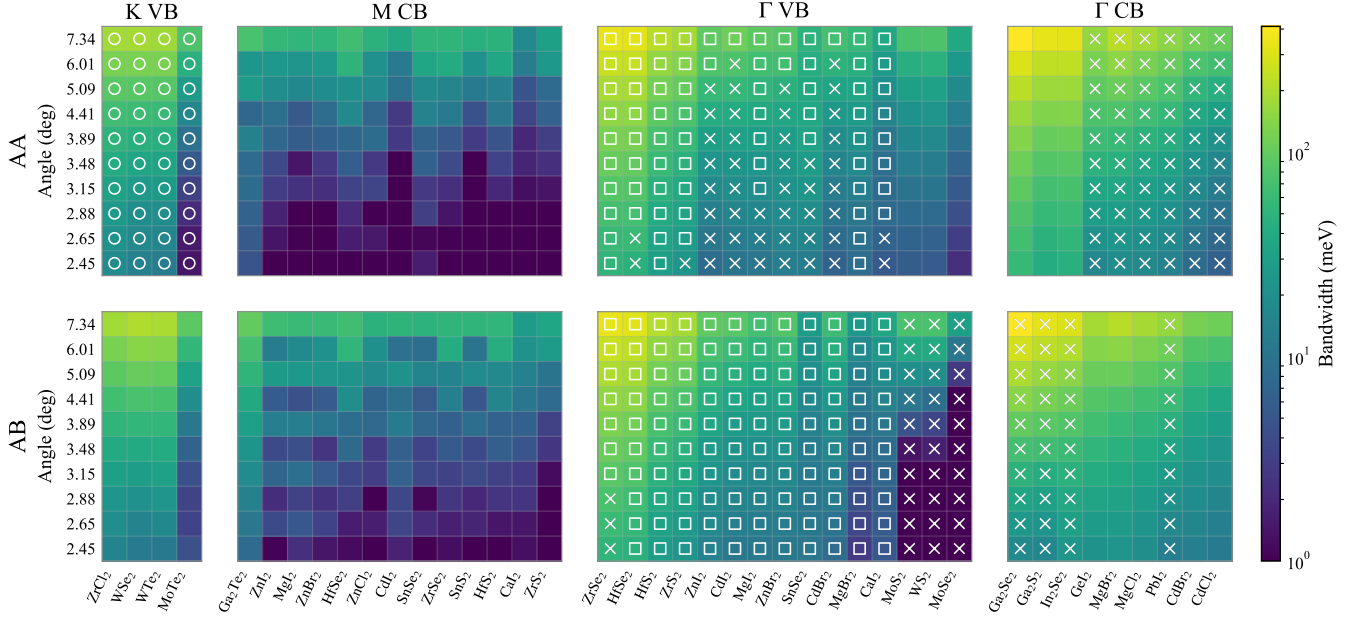


FIG. 2. **Twist-angle dependence of moiré bandwidth and topological characteristics.** Heatmaps of the bandwidth $W(\theta)$ are presented for various material classes categorized by valley character (Γ , K , and M) and band-edge origin. Top and bottom panels display results for the AA- and AB-stacking configurations, respectively. The labels VB and CB represent the moiré minibands derived from the parent valence and conduction bands. Color scales encode $W(\theta)$ from the dispersive to the flat regimes. Topological properties are indicated by symbols: squares denote $\mathbb{Z}_2 = 1$ for Γ -valley systems, circles denote $|C_K| = 1$ for K -valley systems, and crosses mark topologically trivial phases. Notably, the absence of symbols identifies minibands that either lack energetic isolation in the K - and Γ -valleys or host sophisticated M -valley topology. Governed by emergent non-symmorphic symmetries, these M -point systems transcend standard topological classifications and manifest unique kagome or quasi-one-dimensional physics [49, 50].

ciples governing emergent quantum states in semiconductor moiré systems.

RESULTS

Leveraging this framework, we construct an extensive twisted 2D material database by systematically screening 43 experimentally realized monolayers, comprising 39 hexagonal and 4 square lattices. The exhaustive enumeration of symmetry-inequivalent stacking configurations yields 91 bilayer prototypes, whose analysis generates over 1,000 moiré band structures.

Valley classification. To distill universal organizing principles from our expansive dataset, we first categorize the triangular-lattice bilayers, which form the largest class in our dataset, based on the valley character of their constituent monolayer band extrema. This classification distinguishes high-symmetry valleys (Γ , K , and M) from off-symmetry Q valleys. While systems with Q -valley extrema manifest intricate twist-angle dependence, which is driven by the sensitive interplay of moiré band folding and inter-valley hybridization, symmetry-anchored valleys exhibit more systematic and class-dependent behaviors. Consequently, our analysis prioritizes these

high-symmetry valley systems, as they provide a robust platform for identifying the fundamental rules that govern bandwidth scaling and topological reconstruction in moiré materials.

Universal scaling and valley-controlled hierarchy. The global twist-angle dependence of moiré bandwidths, visualized as a spectral heatmap in Fig. 2, reveals a well-defined hierarchy dictated by the valley character of the parent electronic states. A central finding is that the evolution of moiré minibands hinges not merely on the high-symmetry momentum of the parent valley, but primarily on the specific valley hosting the low-energy states, which leads to qualitatively distinct regimes of twist-angle dependence. Among these, Γ -valley systems exhibit the most systematic behavior, where the bandwidth hierarchy across different compounds is largely preserved upon twisting. For most Γ -valley systems, the bandwidth diminishes smoothly and follows a characteristic quadratic scaling, $W(\theta) \propto \theta^2$, in the small-angle limit, consistent with a folding-dominated regime. Nevertheless, stacking-specific interactions can trigger significant departures from this trend, particularly in AB-stacked transition metal dichalcogenides such as MoTe_2 , where the bandwidth quenches more precipitously [22]. Such deviations signal a transition to a strong-coupling regime,

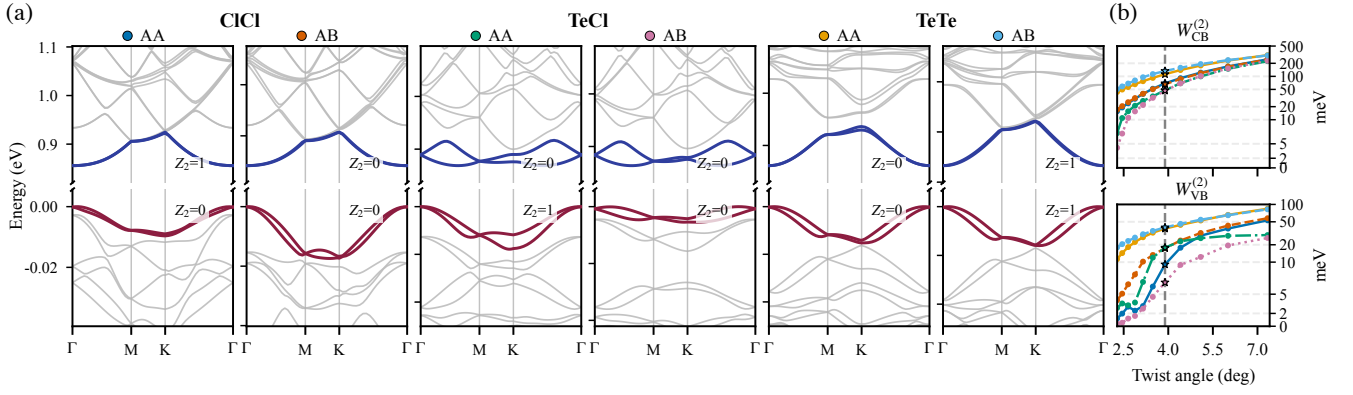


FIG. 3. **Stacking-dependent moiré band structures in BiTeCl bilayer prototypes.** (a) Electronic band structures at a twist angle of 3.89° for representative prototypes, with the $\mathbb{Z}_2$ invariant of the two bands closest to the Fermi level indicated. (b) Twist-angle evolution of the corresponding conduction- and valence-band bandwidths.

in which the large carrier effective mass and the depth of the moiré potential cooperatively drive the electronic states toward the atomic limit. Notably, this accelerated quenching is also observed in K -valley AB configurations, where spin-prohibited tunneling necessitates a regime of strong potential-induced localization that suppresses the kinetic energy more effectively than in the AA-stacked counterparts [22, 25]. In contrast, M -valley systems lack such systematic scaling, as their bandwidth evolution manifests a strong material dependence without a discernible universal trend.

Twist-induced topology from valley structure. The valley-dependent hierarchy that governs bandwidth scaling also dictates the nature and robustness of topological minibands (Fig. 2). In Γ -valley systems, twisting quenches the kinetic energy, facilitating the formation of well-isolated minibands that are highly susceptible to gap opening and topological reconstruction. The resulting topology is fundamentally dictated by the stacking configuration of the parent untwisted bilayer, a dependence most prominently manifested at the band edges. Specifically, at the valence band maximum (VBM) of systems like CdI_2 and HfSe_2 , one configuration sustains a robust $\mathbb{Z}_2 = 1$ insulating state, whereas another undergoes a transition to a trivial phase ($\mathbb{Z}_2 = 0$) in the small-angle moiré regime. Conversely, at the VBM of transition metal dichalcogenides (TMDs) such as MoSe_2 , AB-stacked moiré structures consistently host a trivial topology, while the alternative AA stackings exhibit a vanishingly small gap that mimics the quintessential gapless Kane-Mele model [55]. A strikingly parallel stacking dependence is mirrored at the conduction band minimum (CBM), where one configuration is likewise trivial, while the other similarly manifests the gapless Kane-Mele physics. K -valley systems offer a complementary realization of symmetry-governed topology, where parent Berry curvature is redistributed among moiré minibands. In parallel (AA) stacked bilayers, spin-aligned in-

terlayer tunneling facilitates coherent hybridization, lifting the valley degeneracy and yielding well-isolated minibands with nonzero valley Chern numbers ($|C_K| = 1$). Conversely, antiparallel (AB) stacking preserves inversion symmetry, which, together with spin-valley locking, enforces band degeneracy at Γ and typically precludes the isolation of topmost minibands [22, 25, 44]. In contrast, M -valley systems host a sophisticated topological landscape governed by emergent non-symmorphic symmetries that transcend standard classifications and manifest unique kagome or quasi-one-dimensional physics [49, 50].

Beyond band topology, our database also resolves the quantum geometry of isolated moiré manifolds. By quantifying the relative fluctuation δ_g of the quantum metric $\text{Tr } g(\mathbf{k})$, as detailed in Supplementary Section VIII [56], we identify geometrically uniform flat topological bands that are more Landau-level-like and therefore more relevant for fractionalized phases [57–59]. This metric-based screening highlights AA-type MoTe_2 among Chern-band systems and several $\mathbb{Z}_2 = 1$ manifolds, such as AB-stacked BiTeI , MgBr_2 and CaI_2 , as candidates for further many-body investigation.

Surface termination as a microscopic knob for valley engineering. The decisive role of valley character is further elucidated through surface termination and stacking configuration, which provide microscopic control over the momentum-space location of band extrema. A quintessential example is found in the Janus homobilayer BiTeCl , where the intrinsic inequivalence of Te and Cl surfaces allows for three distinct bilayer terminations: Cl–Cl, Te–Cl, and Te–Te. Our analysis reveals that these termination environments fundamentally reconfigure the parent valley structure. Specifically, while the symmetric Te–Te and Cl–Cl configurations retain their valence band maxima at the Γ point, the asymmetric Te–Cl termination shifts the band extrema to off-symmetry Q valleys [56].

The consequences of this valley shift are directly reflected in the moiré spectra at a twist angle of 3.89° (Fig. 3). The Te-Cl termination manifests the complex, strongly hybridized dispersions typical of Q -valley systems, whereas the symmetric terminations yield well-defined minibands dominated by Γ -derived folding. Notably, the Te-Te configuration provides a clear demonstration of stacking-controlled topology, where AA stacking results in a trivial phase ($\mathbb{Z}_2 = 0$), while AB stacking stabilizes a robust $\mathbb{Z}_2 = 1$ phase over a wide twist-angle range. These findings underscore that surface termination and stacking serve as precise experimental knobs to tune the valley character of parent monolayers, thereby enabling the targeted engineering of moiré band reconstruction and emergent topological states.

Complementing these representative examples, we provide a comprehensive dataset in the Supplementary Information [56], encompassing moiré band structures, gap statistics, and topological characterizations for all investigated materials. To support community-driven discovery, we have developed an interactive online database that enables seamless exploration of this extensive materials library. For each entry in the repository, the platform delivers a detailed profile of the electronic structure and topology, including Wilson loop spectra, Berry curvature maps, and quantum geometry, systematically cross-referenced with the properties of the corresponding parent monolayers. This integrated framework facilitates a direct evaluation of the interplay between moiré-induced effects and intrinsic material characteristics, establishing a robust foundation for the design and experimental verification of twisted quantum matter.

DISCUSSION

In summary, we have established a scalable framework for the systematic exploration of twisted moiré semiconductors, uncovering a unified physical landscape that governs their electronic and topological properties across diverse material classes. Our extensive, angle-resolved database reveals that the emergence of moiré-induced phenomena is fundamentally dictated by the interplay between valley character and stacking symmetry, which together establish a well-defined hierarchy of bandwidth renormalization and topological reconstruction. Specifically, Γ -valley systems manifest a near-universal quadratic scaling of bandwidth with twist angle and host robust $\mathbb{Z}_2 = 1$ insulating phases over broad, experimentally accessible regimes. In contrast, K -valley systems exhibit valley-resolved topology inherited from the underlying electronic structure, although the manifestation of these phases remains sensitively dependent on the stacking-mediated inversion symmetry. Whereas M -valley systems display predominantly idiosyncratic behaviors, characterized by irregular bandwidth evolu-

tion and not captured by the standard isolated-band $\mathbb{Z}_2$ /Chern classification, the systematic regularity observed in other valleys provides a predictive framework for the rational design of moiré matter. Collectively, these findings demonstrate that interlayer twisting, modulated by valley degrees of freedom, serves as a universal mechanism for generating emergent quantum states, laying the foundation for a predictive taxonomy and the targeted engineering of two-dimensional twistrionics.

DATA AVAILABILITY

Data supporting the findings of this study are available within the Supplementary Information [56] and are also publicly accessible at the web-based repository: <https://materials.galaxy.iphy.ac.cn/moire>. The dataset includes relaxed moiré structures, electronic band structures, and derived topological quantities such as topological invariants, Wilson-loop spectra, and momentum-resolved Berry curvature and quantum geometry in the Brillouin zone, as well as associated metadata from the twisted materials database.

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CONTRIBUTIONS

Q. Wu conceived the study and co-supervised the project with H. Weng. J. Li, Y. Zhang, and J. Liu developed the computational framework and constructed the twisted moiré materials database. J. Li screened candidate materials. J. Liu carried out structural relaxations, while Y. Zhang performed first-principles calculations and calculated the electronic and topological properties of large-scale moiré structures. Y. Zhang and J. Liu also developed the database website. Y. Cai and T. Zhu assisted with HPC computing and website deployment. J. Li drafted the manuscript with input from all

authors. J. Li, Y. Zhang, and J. Liu wrote the Methods section and the Supplementary Information. All authors discussed the results and commented on the manuscript.

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